

Shawn D. Ryan

Curriculum Vitae

Present Address
2121 Euclid Ave. RT 1538
Cleveland State University
Cleveland, OH 44115
☎ 1-(216)-687-4707

✉ s.d.ryan@csuohio.edu
academic.csuohio.edu/ryan_s

Research Interests

- Mathematical Analysis, Simulations, and Modeling with Apps in Biology and Materials Science
- Applied and Computational Mathematics
- Mathematical Biology
- Collective Motion, Phase Transitions, and Pattern Formation in Active Biosystems/Biomaterials
- Scientific Machine Learning (PINNs and BINNs), Equation Learning

Employment

- 2021 - **Associate Professor**, *Department of Mathematics and Statistics, Cleveland State University, Cleveland, OH*
- 2020 - **co-Director**, *Center for Applied Data Analysis and Modeling (ADAM), Cleveland State University, Cleveland, OH*
- 2020 - **Affiliated Faculty Member**, *Dept. of Chemical and Biomedical Engineering, Cleveland, OH (Courtesy Appointment)*
- 2016 - 2021 **Assistant Professor**, *Department of Mathematics and Statistics, Cleveland State University, Cleveland, OH*
- 2014 - 2016 **Postdoctoral Research Scholar**, *Department of Mathematical Sciences and Liquid Crystal Institute, Kent State University, Kent, OH*
Mentors: Prof. Peter Palfy-Muhoray and Assoc. Prof. Xiaoyu Zheng
- 2009 - 2014 **Graduate Fellow/Teaching Associate**, *Department of Mathematics, Pennsylvania State University, University Park, PA*
Advisor: Prof. Leonid Berlyand
- May - Aug. '10,'11,'12 **Research Aide**, *Materials Science Division, Argonne National Lab, Lemont, IL*
Supervisor: Prof. Igor S. Aronson

Education

- 2009-2014 **Ph.D. in Mathematics**, *Pennsylvania State University, University Park, PA*
Advisor: Professor Leonid Berlyand
Thesis: "Effective Properties and Collective Dynamics in Bacterial Suspensions"
Committee: L. Berlyand (Chair), R. Colby (Materials Science), X. Li, A. Mazzucato
Penn State University Pritchard Dissertation Award
- 2007-2009 **M.S. in Applied Mathematics**, *The University of Akron, Akron, OH*
Advisor: Professor D. Golovaty and Professor J. P. Wilber
Thesis: "Bifurcation and Boundary Layer Analysis for Graphene Sheets"
- 2005-2009 **Honors B.S. in Applied Mathematics, summa cum laude**, *The University of Akron, Akron, OH*
Honors Project: "Boundary Layer Analysis for Graphene Sheets"

Publications

* Indicates Corresponding Author, **CSU Student in Ryan Research Group**, **non-CSU Student**

1. **S. Guha**, **S. D. Ryan***, B. Karamched*, "Macroscopic Dominance from Microscopic Extremes: Symmetry Breaking in Foraging Populations", *Accepted to Europhysics Letters (EPL)* (2026). Preprint on arXiv: <https://arxiv.org/pdf/2603.10911>
2. **C. Gibson**, **S. D. Ryan***, "Toe Twitch or not Toe Twitch: A Frog Pursuit Model", *Journal of Herpetology* **60**(2): 1-5 (2026).
3. **V. Ciluveru**, **J. Simon**, J. M. Farber, **S. D. Ryan**, C. Kothapalli, D. S. Munther* "Peracetic Acid Efficacy and Decay Kinetics in Poultry Processing under Chiller Conditions", *International Journal for Food Microbiology* **448** 111572 (2026). Preprint on BioArXiv: <https://www.biorxiv.org/content/10.1101/2025.09.04.674287.full.pdf>
4. **C. Brown**, M. Potomkin*, **S. D. Ryan**, "Boundary accumulation of active rods in microchannels with elliptical cross-section", *Phys. Rev. E* **112** 045407 (2025).
5. D. Munther*, **S. D. Ryan**, C. Kothapalli, **N. Zekaj**, "Spatial cross-contamination profiles: Effect of water reuse on dynamics of *Campylobacter* and *E. coli* during poultry processing", *Food and Bioprocess Technology* **153**, 472-485 (2025).
6. D. Munther*, **S. D. Ryan**, C. Kothapalli, **N. Zekaj**, "Spatial characterizations of bacterial dynamics for food safety: Modeling for shared water processing environments", *Applied Mathematical Modelling* **139**, 115818 (2025). Preprint on BioArXiv: <https://www.biorxiv.org/content/10.1101/2024.11.04.621859v1>
7. **S. Hartman**, **S. D. Ryan***, B. Karamched*, "Walk This Way: Modeling Foraging Ant Dynamics in Multiple Food Source Environments", *Journal of Mathematical Biology* **89**(41) (2024). Preprint on BioArXiv: <https://doi.org/10.1101/2024.01.20.576461>
8. **A. Chriss**, V. Boerner*, **S. D. Ryan*** "Agent-based modeling of nuclear chromosome ensemble identifies determinants of homolog pairing during meiosis", *PLoS Computational Biology* **20**(5), e1011416 (2024). Preprint on BioArXiv: <https://doi.org/10.1101/2023.08.09.552574>
9. O. Baloglu*, B. S. Marino, S. Q. Latifi, A. Morca, D. S. Munther, **S. D. Ryan** "External Validation of a Clinical Mathematical Model Estimating Postoperative Urine Output Following Cardiac Surgery in Children", *Pediatric Nephrology* 1-6 (2024).
10. **S. D. Ryan*** "BOOK REVIEW: Mathematical Modeling in Biology, A Research Methods Approach", *SIREV: SIAM Review*, Editor: A. T. Layton, **66**(4): 795-805 (2024).
11. A. Tandon*, S. Bhattacharya, O. T. Inan, J. Lasz, S. Latifi, N. Lu, B. Marino, A. Morca, D. Munther, **S. D. Ryan**, O. Baloglu* "Noninvasive cardiac output monitoring in congenital heart disease", *Current Treatment Options in Pediatrics* **9** (4), 247-259 (2023).
12. **N. Zekaj**, **S. D. Ryan**, A. Resnick* "Fluid-Structure Interaction Modelling of Neighboring Tubes with Primary cilium Analysis", *Mathematical Biosciences and Engineering* **20**(2) 3677-369 (2023).
13. **R. Godin**, B. Karamched*, **S. D. Ryan*** "The Space Between Us: Modeling Spatial Heterogeneity in Synthetic Microbial Consortia Dynamics", *Biophysical Reports* **2**(4) 100085 (2022).
14. O. Baloglu*, **S. D. Ryan**, A. Onder, D. Rosen, C. J. Mullett, D. Munther "A Clinical Mathematical Model Estimating Postoperative Urine Output in Children Underwent Cardiopulmonary By-pass for Congenital Heart Surgery", *Journal of Pediatric Intensive Care* (2022).
15. **S. D. Ryan***, T. M. Nawalaniec "An OER Approach to Linear Algebra", *PRIMUS* **32**(6) 721-737 (2022).
16. **M. Kalil**, **N. Baumgartner**, **M. Issa**, **S. D. Ryan**, C. Wirth* "Influence of PEG on the Clustering of Active Janus Colloid" *Colloids and Surfaces A: Physicochemical and Engineering Aspects* **627** 127191 (2021).
17. S. Peled, **S. D. Ryan**, S. Heidenreich, M. Bär, G. Ariel*, A. Be'er "Heterogeneous Bacterial Swarms with Mixed Lengths", *Phys. Rev. E* **103** 032413 (2021). Preprint: <https://arxiv.org/pdf/2011.12612.pdf>
18. **S. D. Ryan**, **Z. McCarthy**, M. Potomkin* "Motor protein transport along inhomogeneous microtubules",

Bulletin of Mathematical Biology **83**(9) 1-29 (2021).

19. **N. Baumgartner**, **S. D. Ryan*** “Model for Ant-Crab Interaction During Migration on Christmas Island”, *Mathematical Biosciences* **330** 108486 (2020). DOI: <https://doi.org/10.1016/j.mbs.2020.108486>
20. **N. Barron**, **S. D. Ryan**, T. Heus* “Reconciling chord length distributions and area distributions for fields of fractal cumulus clouds” *Atmosphere* **11**(8) 824 (2020). Open Access Publication, DOI: [10.3390/atmos11080824](https://doi.org/10.3390/atmos11080824)
21. **A. Sulak**, **W. Calabrese**, **S. D. Ryan**, T. Heus* “The contributions of shear and turbulence to cloud overlap for cumulus clouds” *Journal of Geophysical Research: Atmospheres* **125** (2020)., DOI: [10.1029/2019JD032017](https://doi.org/10.1029/2019JD032017)
22. **N. Praljak**, **S. D. Ryan**, A. Resnick* Pulsatile Flow Through Idealized Renal Tubules: Fluid-Structure Interaction and Dynamic Pathologies, *Mathematical Biosciences and Engineering*, **17**(2) 1787-1807 (2020). Open Access Publication, DOI: [10.3934/mbe.2020094](https://doi.org/10.3934/mbe.2020094)
23. **S. D. Ryan***, “Role of hydrodynamic interactions in chemotaxis of bacterial populations”, *Physical Biology*, **17** 016003 (2020).
 ○ Chosen as featured article on Phys. Biology Homepage 2020
24. **M. Issa**, **N. Baumgartner**, **M. A. Kalil**, **S. D. Ryan**, C. Wirth*, “Charged Nanoparticles Quench the Propulsion of Active Janus Colloids”, *American Chemical Society Omega* **4**(8) 13034-13041 (2019). Open Access Publication, DOI: [10.1021/acsomega.9b00765](https://doi.org/10.1021/acsomega.9b00765)
25. **Z. McCarthy***, B. Smith, A. Fazil, J. Wu, **S. D. Ryan**, D. Munther, “An individual-carcass model for quantifying bacterial cross-contamination in an industrial three-stage poultry scalding tank”, *Journal of Food Engineering* **262** 142-153 (2019).
26. G. Ariel, M. Sidortsov, **S. D. Ryan**, S. Heidenreich, M. Bar, A. Be’er* “Collective dynamics of two-dimensional swimming bacteria: Experiments and models”, *Phys. Rev. E* **98** 032415 (2018).
27. L. Berlyand, V. Mityushev, and **S. D. Ryan***, “Effect of randomness on the distribution of multiple Ginzburg-Landau vortices pinned by small holes”, *IMA Journal of Applied Mathematics* **83**(6) 977-1006 (2018).
28. **Z. McCarthy***, B. Smith, A. Fazil, J. Wu, **S. D. Ryan**, D. Munther, “pH dependent C. jejuni thermal inactivation models and application to poultry scalding”, *Journal of Food Engineering* **223** 1-9 (2018).
29. **A. Sulak**, **S. D. Ryan*** “Optimal shape of water towers”, *SIAM Undergraduate Research Journal (SIURO)* **10** 233-247 (2017).
30. **Z. McCarthy**, B. Smith, A. Fazil, J. Wu, **S. D. Ryan**, D. Munther*, “Individual based modeling and analysis of pathogen levels in poultry chilling process”, *Mathematical Biosciences* **294** 172-180 (2017).
31. **S. D. Ryan**, G. Richards, X. Zheng, P. Palffy-Muhoray*, “A finite volume method for computing flow induced orientation of nematic liquid crystals”, *Molecular Crystals and Liquid Crystals (MCLC)* **647**:(1) 207-215 (2017).
32. **Z. McCarthy**, D. Munther*, **S. D. Ryan***, J. Wu, “Mechanisms of bacterial contamination in poultry chill tanks”, *Technical Report for the Laboratory of Foodborne Zoonoses, Public Health Agency of Canada* (2017).
33. **S. D. Ryan**, G. Ariel, A. Be’er*, “Anomalous fluctuations in the orientation and velocity of swarming bacteria”, *Biophysical Journal*, **111**(1), 247-255 (2016). <http://dx.doi.org/10.1016/j.bpj.2016.05.043>
34. **S. D. Ryan***, X. Zheng, P. Palffy-Muhoray, “Curvature-driven foam coarsening on the sphere: A computer simulation”, *Physical Review E*, **93**, 053301 (2016). doi:10.1103/PhysRevE.93.053301
 ○ Chosen for Phys. Rev. E Kaleidoscope (best aesthetic images)
35. M. Potomkin*, **S. D. Ryan**, and L. Berlyand, “Effective rheological properties in semidilute bacterial suspensions”, *Bulletin of Mathematical Biology*, **78**(3), 580-615,(2016). doi:10.1007/

s11538-016-0156-2

36. **S. D. Ryan*** "A model for collective dynamics in ant raids", *Journal of Mathematical Biology*, **72**(6), 1579-1606, (2016) doi:10.1007/s00285-015-0929-5.
37. **S. D. Ryan**, V. Mityushev, V. M. Vinokur*, and L. Berlyand "Rayleigh approximation for ground states of the Bose and Coulomb glasses", *Nature Scientific Reports* **5**, 7821 (2015).
○ Press: **Mathematical approach provides a new step in resolving mystery of glass**
38. **S. D. Ryan***, "Effective Properties and Collective Dynamics in Bacterial Suspensions", *Ph.D. Dissertation*, The Pennsylvania State University, (2014).
39. **S. D. Ryan**, A. Sokolov, L. Berlyand, and I. S. Aranson*, "Correlation properties of collective motion in bacterial suspensions", *New Journal of Physics* **15**, 105021 (2013).
○ Chosen as a **"Highlight of 2013" for Soft Matter and Biophysics**
40. **S. D. Ryan***, L. Berlyand, B. M. Haines, and D. A. Karpeev, "A Kinetic Model of Semi-Dilute Bacterial Suspensions", *SIAM Multiscale Modeling and Simulation* **11**(4), 1176-1196 (2013).
41. **S. D. Ryan**, D. Golovaty, and J. P. Wilber*, "Buckling of a Graphene Sheet Perpendicular to a Rigid Substrate", *International Journal of Solids and Structures* **49**, 3681-3692 (2012).
42. **S. D. Ryan**, B. M. Haines, L. Berlyand, F. Ziebert, and I. S. Aranson*, "Viscosity of Bacterial Suspensions: Hydrodynamic Interactions and Self-Induced Noise", *Rapid Comm. Phys. Rev. E* **83**, 050904R (2011).
43. **S. D. Ryan***, "Bifurcation and Boundary Layer Analysis for Graphene Sheets", *Ohiolink*, Master of Science Thesis at the University of Akron (2009).
44. J. Galagher, Y. Milman, **S. Ryan**, D. Golovaty*, J. P. Wilber*, and A. Buldum, "A Buckling Problem for Graphene Sheets", *Proceedings of the 11th International Congress on Continuum Modeling of Discrete Systems*, Les Presses de École des Mines de Paris (2007).

Submitted Papers

45. **A. Brown**, D. Munther, **S. D. Ryan*** "Modeling Multi-Phase Growth in Cyanobacterial Harmful Algal Blooms", *Submitted* (2026).

Preprints / In Progress

46. **V. Ciluveru**, D. Munther, **S. D. Ryan**, C. Kothapalli "Free Chlorine Decay Kinetics and Efficacy in Pathogen Reduction During Poultry Processing", *In Preparation*.
47. **V. Ciluveru**, D. Munther, **S. D. Ryan**, C. Kothapalli "Cross-contamination Dynamics of Salmonella in Scalding and Chilling Processes", *In Preparation*.
48. **J. Palos**, **C. Brown**, M. Potomkin*, **S. D. Ryan*** "Contamination Dynamics of Active Rods in Long Micro-channels", *In Preparation*.
49. **C. Mark**, **S. D. Ryan***, B. Karamched*, "Model for Branching in Ant Trail Formation", *In Preparation*.

Fellowships, Honors, and Awards

- | | |
|-----------|---|
| 2025 | Named Editorial Board Member (Review Editor) of <i>Frontiers in Systems Biology</i> |
| 2025 | Named Editorial Board Member (Associate Editor) of <i>Involve: A Journal of Mathematics</i> |
| 2025 | Invited Presentation to CSU Board of Trustees, Presentation of Accomplishments During Successful Sabbatical in Spring 2024 |
| 2025,2022 | CSU Golden Apple Teaching Award , CSU Alumni Association, Recognizes |
| 2020,2018 | the contributions of outstanding faculty/staff members on the lives of students |
| 2023-2024 | Provost's Faculty Merit Award , For Outstanding Teaching and Research |

- 2023-2024 [CSU Distinguished Faculty Award for Teaching](#), *For Outstanding Teaching and Student Mentorship, two awards for tenured/tenure-track over whole university*
- 2023-2024 [Jearl D. Walker Outstanding Teaching Award](#), *College of Arts and Sciences*
- 2022-2023 **Provost's Faculty Merit Award**, *For Outstanding Teaching*
- 2021-2022 **Faculty Champion of Teaching in OneNote Award**, *CSU Center for Faculty Excellence and IS&T*
- 2021 **Letter of Recognition for Work on OER and Affordable Learning Resources**, *CSU OER Committee and Affordable Learning Team*
- 2020-2021 **Faculty Innovator Award**, *CSU Center for Faculty Excellence*, Invited to present a webinar on keeping students engaged during remote learning.
- 2019-2020 **Provost's Faculty Merit Award**, *For Outstanding Teaching and Research*
- 2019 **Featured Manuscript on J. Physical Biology Homepage**, *Given free open access for two months*
- 2017-2018 **Provost's Faculty Merit Award**, *For Outstanding Teaching and Research*
- 2017-2018 **Cleveland State Teaching Enhancement Award (TEA) Grant**
- 2017-Present **Level 1 University Graduate Faculty**, *Cleveland State University*
- 2016 **Kent State Department of Athletics Faculty Honor**, *Kent State University*, *For being instrumental in student-athlete academic success*
- 2014-2015 [Certificate for "Dedicated Service and Outstanding Contributions toward Student Success"](#), *KSU Division of Diversity, Equity, and Inclusion Upward Bound*
- 2014 [Pritchard Dissertation Award](#), *Outstanding dissertation work in Mathematics*
- 2014 **Nomination for Alumni Dissertation Award**, *Outstanding dissertation work at Penn State University*
- 2013 **Manuscript receives "Highlight of 2013" Award**, *New Journal of Physics*
- 2012 **Department of Mathematics Teaching Award**, *Pennsylvania State University*
- Nominated for PSU Department of Mathematics Teaching Award in 2011, 2012, and 2013
- 2011 **Promotion to Graduate Teaching Associate**, *Department of Mathematics, Pennsylvania State University*
- 2011 **Teaching With Technology Certificate**, *Graduate School, Penn State University*
- 2009-2014 [University Graduate Research Fellowship](#), *Pennsylvania State University*
- 2009 **August and Ruth Homeyer Graduate Fellowship**, *Pennsylvania State University*
- 2009 **University of Akron Outstanding Senior**, *Top 10 graduating seniors at U. Akron*
- 2008 [Barry M. Goldwater Scholar](#), *Outstanding undergraduate research in the STEM fields. First person at the University of Akron ever to win*
- 2007 **Alberta M. and William C. King Memorial Scholarship for Outstanding Jr./Sr. in Mathematics**, *The University of Akron*
- 2006 **Dr. Mary E. Maxwell Scholarship for Outstanding Freshman in Mathematics**, *The University of Akron*

2005 **University of Akron Honors Scholar of Excellence**, Awarded to top incoming undergraduates of the University of Akron Honors College

Funded Grants

External

- 2023-2026 **USDA NIFA Grant #2023-67017-39208**, *Improved pathogen control for poultry processing: Experimentally-validated mathematical models for scalding, chilling, and post-chilling*, co-PI w/ PI D. Munther and co-PI C. Kothapalli, \$498,177.60
- 2023-2026 **NSF REU Grant**, *REU: Synthesis, Assembly and Characterization of Soft Matter Systems*, Senior Personnel under PIs K. Streletzky and J. Bickel (Funded Jan 2023, NCE to start in 2024), \$365,000
- 2017-Present **Ohio Supercomputing Center Grant**, *Start-Up Computational Hours*, PI Grant #PFS0230-1
- 2017 **Public Health Agency of Canada Grant**, *Mechanisms of Bacterial Contamination in Poultry Chill Tanks*, Co-PI (with D. Munther) Grant #582090, Funded Jan. 2017-Mar. 2017
- 2015 **SIAM Young Researcher Travel Grant**, *For ICIAM 2015, Beijing, China. (Invited Minisymposium Speaker)* Award Amount: \$2000
- 2015 **Elsevier Mathematics Sponsorship Fund**, Award Amount: \$1500
- 2014 **Society of Industrial and Applied Mathematics (SIAM) Travel Grant**, *SIAM Conference on the Life Sciences 2014. (Invited Minisymposium Speaker)* Award Amount: \$650

Internal

- 2026 **CSU USRA Grant**, *USRA: Reducing pathogen cross-contamination during poultry processing: modeling and analysis*, co-PI (with PI D. Munther and co-PI C. Kothapalli, Submitted for Summer 2026)
\$5000
- 2026 **CSU USRA Grant**, *USRA: Inactivation of Salmonella During Poultry Scalding: efficacy of pH and temperature under simulated commercial conditions*, co-PI (with co-PI D. Munther and PI C. Kothapalli, Submitted for Summer 2026)
\$5000
- 2025 **CSU USRA Grant**, *USRA: PH-Dependent Thermal Inactivation Models for Pathogen Control During Poultry Scalding*, co-PI (with PI D. Munther and co-PI C. Kothapalli, Summer 2025)
\$4750
- 2025 **CSU Grant**, *A& S Summer Stipend for RSCA*, For mentoring CSU students during summer
\$500
- 2024 **CSU USRA Grant**, *USRA: Experimentally Validated Mathematical Models for Pathogen Control in Food Industry*, co-PI (with PI D. Munther, Submitted for Summer 2024)
\$4410
- 2022 **CSU USRA Grant**, *Evaluating the mixing and precipitation of clouds as a function of size*, co-PI (with PI T. Heus, Funded for Summer 2022)

- 2021 **CSU Faculty Research Development Grant**, *Modeling tools to reduce pathogen cross-contamination and enhance compliance for poultry chilling*, co-PI (w/ PI D. Munther and co-PI C. Kothapalli)
- 2020 **CSU USRA Grant**, *Evaluating the mixing and precipitation of clouds as a function of size*, co-PI (with T. Heus, Funded for Summer 2020)
- 2019-2020 **CSU Textbook Affordability Grant**, *Open Text and Module-based learning for Linear Algebra*, PI
- 2019-2020 **Faculty Research Development (FRD) Grant**, *Swimming near deformable surfaces*, PI (with co-PI C. Wirth, funded April 2019)
- 2019 **CSU USRA Grant**, *Influence of interfacial properties on propulsion of active particles near an air/water interface*, co-PI (with C. Wirth, funded March 2019)
- 2019 **CSU USRA Grant**, *Comparing cloud size distributions from Total Sky Imaging observations with Large Eddy Simulations*, co-PI (with T. Heus, funded March 2019)
- 2018 **Cleveland State Undergraduate Summer Research Award**, **Co-PI** (PI C. Wirth)
- 2018 **Cleveland State Undergraduate Summer Research Award**, **Co-PI** (PI T. Heus)
- 2017-2018 **Cleveland State Teaching Enhancement Award (TEA) Grant**, (Awarded Feb. 2017)
- 2017 **Cleveland State Undergraduate Summer Research Award**, **Co-PI** (PI T. Heus)

Grants Under Review

- 2026-2029 **USDA NIFA Grant**, *Experimentally informed predictive bio-mapping of pathogen contamination and control during poultry chilling*, co-PI w/ PI D. Munther, co-PI C. Kothapalli, co-PI E. McMillan (USDA) (Submitted Oct 2025, \$635,000)
NOTE: this grant submission features a new collaboration with microbiologist Dr. Elizabeth McMillan at the USDA National Poultry Research Center.
- 2027-2029 **NSF REU Grant Renewal**, *REU: Synthesis, Assembly and Characterization of Soft Matter Systems*, Senior Personnel under PIs K. Streletzky and J. Bickel, TBD

Conferences/Sessions Organized

- July 2026 **Minisymposium co-Organizer**, *Recent Advances in Mathematical Methods for Understanding Biological Systems*, Society for Industrial and Applied Mathematics Annual Meeting, Cleveland, OH
- July 2022 **Minisymposium co-Organizer**, *Mathematics of Interacting Systems with Applications to Biology and Medicine*, Society for Industrial and Applied Mathematics Annual Meeting, Pittsburgh, PA
- June 2021 **Minisymposium co-Organizer**, *Mathematical modeling of emergent phenomena in cell colonies*, Society for Mathematical Biology Annual Meeting, UC Riverside, CA

- Feb 2020 **International Conference co-Organizer**, *Conference on Mathematics of complex systems in biology and medicine*, Part of the Residential Month of Mathematics Issues in Biology at CIRM/University Aux-Marseille, Marseille, France
- Sept. 2019-2024 **Organizer**, *NSF GRFP and Other Graduate Fellowships Workshops*, 3 Part Series, Office of Research, CSU, Cleveland, OH
- July 2017 **Minisymposium Organizer**, *Kinetic Models with Applications in Biology*, 2017 SIAM Annual Meeting, Pittsburgh, PA
- Apr 2017 **Chair of Scientific/Organization Committee**, *2017 Northeast Ohio Applied Mathematics Forum*, Hosted by Cleveland State University, Cleveland, OH
- Nov 2014 **Co-Organizer (with B. Jaye and S. Nitzan)**, *Upward Bound Math Skills Academy*, Kent State University, Kent, OH
Low income high school students seeking to become first generation college students in STEM fields.

External Presentations

- July 2026 **Invited Minisymposium Speaker**, *Multiscale Modeling in Microbial Consortia: Spatial Heterogeneity and Collective Dynamics*, 2026 SIAM Conference on the Life Sciences, Cleveland, OH
- Sept 2025 **Invited Colloquium Speaker**, *A Mathematical Microscope: Revealing Biological Structure and Dynamics Across Scales*, Department of Biology, Geology, and Environmental Sciences Colloquium, Cleveland State University, Cleveland, OH
- July 2025 **Poster**, *Evaluating Free Chlorine Efficacy in Poultry Processing: Decay Kinetics, Pathogen Reduction, and Organic Matter Interactions*, 2025 Poultry Science Association Annual Meeting, Raleigh, NC
- Nov 2024 **Invited Talk**, *Careers in Math + Biology*, Ellsworth Hill Elementary, Hudson, OH
- July 2024 **Invited Poster**, *Improved pathogen control for poultry processing: Experimentally-validated mathematical models for scalding, chilling, and post-chilling*, USDA-NIFA Annual Meeting for Investigators, Long Beach, CA
- July 2024 **Invited Poster**, *Spatial modeling of the poultry chilling process: impact of water recirculation and counterflow on E. coli and Campylobacter dynamics*, International Assoc. for Food Protection Annual Meeting, Long Beach, CA
- Apr 2024 **Contributed Talk**, *Decoding Biological Complexity with Mathematical Modeling and HPC*, Ohio Supercomputing Center Research Symposium, Columbus, OH
- Mar 2024 **Invited Talk**, *Agent-based Models for Applications in Mathematical Biology*, Applied Math Seminar, Kent State University, Kent, OH
- Mar 2024 **Invited Talk**, *Patterns of Life: Mathematical Exploration of Self-Organization in Biological Systems*, Youngstown State University Mathematics Colloquium, Youngstown, OH
- Feb 2024 **Invited Talk**, *Patterns of Life: Mathematical Exploration of Self-Organization in Biological Systems*, Kent State University Mathematics Colloquium, Kent, OH
- Feb 2024 **Invited Talk**, *Patterns of Life: Mathematical Exploration of Self-Organization in Biological Systems*, University of Pittsburgh Applied Math Seminar, Pittsburgh, PA
- April 2023 **Invited Talk**, *Math Modeling for Self-Organization In the Biosciences*, AMS Central Section Meeting, Cincinnati, OH -Declined due to birth of child

- Mar 2023 **Invited Talk**, *Better Together: Math Gives Novel Insight Into Microscale Biology*, Computational and Applied Mathematics Seminar, Univ. of Wyoming
- Feb 2023 **Invited Talk**, *Modeling identifies chromosome numbers, repulsive and attractive forces as determinants of homolog pairing during meiosis*, Rust Belt Meiosis Meeting, Cleveland, Ohio
- Jul 2022 **Invited Talk**, *Role of Hydrodynamic Interactions in Collective Swimming of Bacteria*, SIAM 2022 Annual Meeting, Pittsburgh, PA
- Feb 2022 **Invited Talk**, *Mathematics Provides Insight into Self-Organization in Biology*, Mathematics Colloquium, Tufts University, Virtual, COVID-19)
- Jan 2022 **Invited Talk**, *Mathematics Provides Insight into Self-Organization in Biology*, Interdisciplinary Center for Quantitative Modeling in Biology Colloquium, University of California-Riverside (Virtual, COVID-19)
- Aug 2021 **Contributed Talk**, *Desmos and Dynamics* (w/ F. Martins and I. Rodrigues), Mathfest 2021, Virtual, COVID-19)
- Jun 2021 **Invited Talk**, *Role of Hydrodynamic Interactions in Collective Swimming of Bacteria*, SMB 2021 Annual Meeting, UC Riverside, CA (Virtual, COVID-19)
- Oct 2020 **Invited Talk**, *Interaction Between Ants and Crabs on Christmas Island*, Structural Biology/Biochemistry Seminar Series, Institute of Molecular Biophysics, Florida State University (Virtual, COVID-19)
- Aug 2020 **Invited Talk**, *Mathematical Modeling for Collective Dynamics in Ant Raids*, Biomath Journal Club - Bertram Group, Florida State University (Virtual)
- Aug 2020 **Invited Poster**, *Role of Hydrodynamics in Chemotaxis of Bacterial Populations*, Soc. Math Biology Annual Meeting, Heidelberg, Germany (Virtual, COVID-19)
- Jul 2020 **Invited Speaker**, *Mathematics Provides Insight Into Self-Organization in Biology*, Applied Math Seminar, U. Waterloo, Canada (Virtual, COVID-19)
- Jul 2020 **Contributed Speaker**, *Role of Hydrodynamics in Chemotaxis of Bacterial Populations*, 2020 SIAM Annual Meeting, Toronto, Can (Virtual, COVID-19)
- Jul 2020 **Invited Speaker**, *Mathematics Provides Insight into Self-Organization in Biology*, Global Seminar on Mathematical Modeling and Applications (GSMMA), Online via Zoom hosted by UMass Amherst and U. Minnesota
- Feb 2020 **Invited Plenary Speaker**, *Mathematics Provides Insight into Self-Organization in Biology*, CIRM Workshop, Marseilles, France
- July 2019 **Contributed Speaker**, *Mathematics Provides Insight into Self-Organization in Biology*, 2019 Society for Mathematical Biology International Meeting, Montreal, Quebec, Canada
- April 2019 **Invited Plenary Speaker**, *Mathematics Provides Insight into Self-Organization in Biology*, Spring 2019 Ohio MAA Section Meeting, The University of Akron, Akron, OH
- Oct 2018 **Invited Minisymposia Speaker**, *Onset of Collective Dynamics in Active Biosystems*, Fall 2018 AMS Central Section Meeting, Ann Arbor, MI
- Aug 2018 **Invited Minisymposia Speaker**, *Onset of Collective Dynamics in Active Biosystems*, 2018 SIAM Life Sciences Meeting, Minneapolis, MN
- Mar 2018 **Invited Minisymposia Speaker**, *Collective Dynamics in Ant Raids*, 2018 AMS Central Section Meeting, The Ohio State University, Columbus, OH

- Feb 2018 **Invited co-author with D. Munther (CSU) (Invited Speaker O. Baloglu (Cle Clinic))**, Mathematical and Computational Modeling of Critical Illness, Grand Rounds, Cleveland Clinic, Cleveland, OH
- Sept 2017 **Invited Minisymposia Speaker**, Kinetic PDE models for active biosystems, 2017 AMS East Section Meeting, State University of New York at Buffalo, Buffalo, NY
- July 2017 **Invited Minisymposia Speaker**, Kinetic Models for Active Biosystems, 2017 SIAM Annual Meeting, Pittsburgh, PA
- May 2017 **Invited Poster Presenter**, Algorithm for studying flow-induced phase transitions in nematic liquid crystals, Non-convexity, non-locality and incompatibility: from materials to biology, University of Pittsburgh, Pittsburgh, PA
- Feb 2017 **Invited Speaker**, Collective Dynamics in Active Biological Systems, SIAM Applied Math Seminar, University of Akron, Akron, OH
- Aug 2016 **Poster Presentation**, Computational algorithm for studying vorticity driven dynamics in nematic LCs, 26th International Liquid Crystal Conference, Kent, OH
- July 2016 **Poster Presentation and Speaker**, Curvature driven foam coarsening on the sphere, Recent Trends and Advances in Nonlinear Analysis, Carnegie Mellon University, Pittsburgh, PA
- Jun 2016 **Participant**, Medical Research: The Right Prescription for Economic Growth, Research!America Workshop, Northeast Ohio Medical University, Rootstown, OH
- Feb 2016 **Invited Speaker**, Collective Dynamics in Active Biosystems, Department of Mathematics Colloquium, Cleveland State University, Cleveland, OH
- Feb 2016 **Invited Speaker**, Curvature driven foam coarsening on the sphere, Applied Mathematics Seminar, Kent State University, Kent, OH
- Jan 2016 **Invited Speaker**, Curvature driven foam coarsening on the sphere, Northeast Ohio Applied Mathematics Workshop, Kent State University, Kent, OH
- Oct 2015 **Invited Speaker**, Can Mathematical Modeling Help Endangered Species?, Choose Ohio First Scholars Seminar, Kent State University, Kent, OH
- Sept 2015 **Invited Speaker**, A Model for Collective Dynamics in Ant Raids, Mathematics of the Life Sciences Seminar, Case Western Reserve University, Cleveland, OH
- Aug 2015 **Invited Minisymposium Speaker**, Collective Dynamics in Active Biological Systems, The International Congress on Industrial and Applied Mathematics (ICIAM) 2015, Beijing, China
- May 2015 **Participant**, KI-Net Workshop: Groups and interactions in data, networks and biology, Carnegie Mellon University, Pittsburgh, PA
- Apr 2015 **Invited Speaker**, A Model for Collective Dynamics in Ant Raids, Applied Math Seminar, The Ohio State University, Columbus, OH
- Apr 2015 **Invited Speaker**, Collective Dynamics in Active Biological Systems, Applied Math Lab Seminar, Courant Institute, New York University, New York, NY
- Feb 2015 **Invited Speaker**, A Model for Collective Dynamics in Ant Raids, Comp. and Appl. Math Seminar, Kent State University, Kent, OH
- Nov 2014 **Invited Speaker**, Collective Motion: Why 2 is better than 1, Choose Ohio First Scholars Seminar, Kent State University, Kent, OH

- Oct 2014 **Invited Speaker**, *Effective Properties and Collective Dynamics in Bacterial Suspensions*, Applied Mathematics Seminar, Ohio State University, Columbus, OH
- Sept 2014 **Invited Speaker**, *Effective Properties and Collective Dynamics in Bacterial Suspensions*, Mathematics of the Life Sciences Seminar, Case Western Reserve University, Cleveland, OH
- Sept 2014 **Invited Speaker**, *Effective Properties and Collective Dynamics in Bacterial Suspensions*, Applied and Computational Mathematics Seminar, Kent State University, Kent, OH
- Aug 2014 **Invited Minisymposium Speaker**, *Collective Effects and Correlations in Semi-dilute Bacterial Suspensions*, SIAM Conference on the Life Sciences, Charlotte, NC
- May 2014 **Invited Speaker**, *Effective Properties and Collective Dynamics in Bacterial Suspensions*, Dissertation Defense, Pennsylvania State University, University Park, PA
- Apr 2014 **Invited Speaker**, *Effective Properties and Collective Dynamics in Bacterial Suspensions*, Applied Mathematics / Liquid Crystal Institute Seminar, Kent State University, Kent, OH
- Oct 2013 **Invited Speaker and Poster Presenter**, *Collective Effects and Correlations in Semi-dilute Bacterial Suspensions*, PDEs and Dynamical Systems in Biology Conference, Bar-Ilan University, Israel
- Jun 2012 **Poster Presenter**, *A Force Dipole Model for Semi-dilute Bacterial Suspensions*, Workshop on Bacteria: Experiments and Modeling, Argonne and Notre Dame University, Chicago, IL
- Jan 2012 **Invited Speaker**, *Coupled PDE/ODE Model for Semi-dilute Bacterial Suspensions*, Oral Comprehensive Exam, Dept. of Mathematics, Penn State University, University Park, PA
- Jan 2012 **Poster Presenter**, *A Force Dipole Model for Semi-dilute Bacterial Suspensions*, MBI CTW: Free Boundary Problems in Biology, Mathematical Biosciences Institute (MBI), The Ohio State University, Columbus, OH
- Jun 2011 **Invited Speaker**, *A Force Dipole Model for Semi-dilute Bacterial Suspensions*, Biomat 2011: Prospective in the Life Sciences, Granada, Spain
- May 2009 **Invited Speaker**, *Bifurcation and Boundary Layer Analysis for Graphene Sheets*, Masters Thesis Oral Defense, Dept. of Mathematics, The University of Akron, Akron, OH
- Mar 2008 **Invited Speaker**, *A Buckling Problem for Graphene Sheets*, Conference on Undergraduate and Graduate Student Research, The University of Akron, Akron, OH
- Jan 2008 **Invited Speaker**, *A Buckling Problem for Graphene Sheets*, 2008 Joint Mathematics Meeting, San Diego, CA
- Nov 2007 **Invited Speaker**, *Modeling Interacting Sheets of Graphene*, Undergraduate Mathematics Day, The University of Dayton, Dayton, OH

Internal Presentations

- Mar 2026 **Invited Speaker**, *Modeling Bacteria: From Individuals to Communities*, Math Club, Cleveland State University, Cleveland, OH

- Mar 2026 **Invited Speaker**, *Agent-based Modeling in Biology*, Junior Seminar Spring 2026, Cleveland State University, Cleveland, OH
- Feb 2026 **Invited Speaker**, *How to Find and Earn Grants as a Grad Student*, School of Graduate Studies Seminar, Cleveland State University, Cleveland, OH
- Mar 2025 **Invited Speaker**, *Agent-based Modeling in Biology*, Junior Seminar Spring 2025, Cleveland State University, Cleveland, OH
- Feb 2025 **Invited Speaker**, *How to Find and Earn Grants as a Grad Student*, School of Graduate Studies Seminar, Cleveland State University, Cleveland, OH
- Feb 2025 **Invited Speaker**, *Mathematics Provides Insight into Biology*, SPS-Choose Ohio First Seminar Spring 2025, Cleveland State University, Cleveland, OH
- Feb 2025 **Invited Speaker**, *Patterns of Life: What Math Can Teach Us About Biology*, CSU Honors College Seminar: Edge of Knowledge, Cleveland State University, Cleveland, OH
- Jan 2025 **Invited Speaker**, *Patterns of Life: Mathematical Exploration of Self-Organization in Biological Systems*, Junior Seminar Spring 2025, Cleveland State University, Cleveland, OH
- Jan 2025 **Invited Speaker**, *PLOA Accomplishments from Spring 2024*, CSU Board of Trustees Meeting, Cleveland State University, Cleveland, OH
- Sept 2024 **Invited Speaker**, *Preparing for the NSF GRFP (3-Part Workshop)*, CSU NSF GRFP Workshop, Cleveland State University, Cleveland, OH
- Sept 2024 **Invited Poster**, *PAA decay modeling at different steps of poultry processing*, Fall 2024 USRA Poster Session, Cleveland State University, Cleveland, OH
- Jan 2024 **Invited Speaker**, *Patterns of Life: Mathematical Exploration of Self-Organization in Biological Systems*, Junior Seminar Spring 2024, Cleveland State University, Cleveland, OH
- Nov 2023 **Invited Speaker**, *Mathematics Provides Insight into Biology*, SPS-Choose Ohio First Seminar Fall 2023, Cleveland State University, Cleveland, OH
- Sept 2023 **Invited Speaker**, *Preparing for the NSF GRFP (3-Part Workshop)*, CSU NSF GRFP Workshop, Cleveland State University, Cleveland, OH
- Sept 2023 **Invited Speaker**, *Mathematics Provides Insight into Biology*, Junior Seminar Fall 2023, Cleveland State University, Cleveland, OH
- Mar 2023 **Invited Speaker**, *How to Find and Earn Grants as a Grad Student*, School of Graduate Studies Seminar, Cleveland State University, Cleveland, OH
- Jan 2023 **Invited Speaker**, *Mathematics Provides Insight into Biology*, Junior Seminar Spring 2023, Cleveland State University, Cleveland, OH
- Nov 2022 **Invited Speaker**, *Mathematics Provides Insight into Biology*, SPS-Choose Ohio First Seminar Fall 2022, Cleveland State University, Cleveland, OH
- Oct 2022 **Invited Speaker**, *Mathematics Provides Insight into Biology*, Junior Seminar Fall 2022, Cleveland State University, Cleveland, OH
- Sept 2022 **Invited Speaker**, *Preparing for the NSF GRFP (3-Part Workshop)*, CSU NSF GRFP Workshop, Cleveland State University, Cleveland, OH
- Aug 2022 **Invited Speaker**, *How To Do Great Research and Scholarship Pt. 2*, CSU USRA Student Seminar, Cleveland State University, Cleveland, OH

- Aug 2022 **Invited Speaker**, *How To Do Great Research and Scholarship Pt. 1*, CSU USRA Student Seminar, Cleveland State University, Cleveland, OH
- Mar 2022 **Invited Speaker**, *Mathematics Provides Insight into Biology*, Junior Seminar Spring 2022, Cleveland State University, Cleveland, OH
- Sept 2021 **Invited Speaker**, *Mathematics Provides Insight into Self-Organization in Biology*, Junior Seminar Fall 2021, Cleveland State University, Cleveland, OH
- Sept 2021 **Invited Speaker**, *NSF GRFP: Research Statement*, NSF GRFP Workshop, CSU Graduate School, Cleveland State University, Cleveland, OH
- Sept 2021 **Invited Speaker**, *NSF GRFP: Personal Statement*, NSF GRFP Workshop, CSU Graduate School, Cleveland State University, Cleveland, OH
- Sept 2021 **Invited Speaker**, *What is the NSF GRFP?*, NSF GRFP Workshop, CSU Graduate School, Cleveland State University, Cleveland, OH
- Aug 2021 **Invited Speaker**, *Post Pandemic Teaching Community*, Gateway Courses Seminar Fall 2021, Cleveland State University, Cleveland, OH
- Jun 2021 **Invited Speaker**, *How to Frame Your Research for the NSF GRFP Program*, RE@CSU Summer Research Experience, Cleveland State University, Cleveland, OH
- Jan 2021 **Invited Speaker**, *Mathematics Provides Insight into Self-Organization in Biology*, Junior Seminar Spring 2021, Cleveland State University, Cleveland, OH
- Oct 2020 **Invited Speaker**, *Mathematics Provides Insight into Biology*, SPS/Choose Ohio First Seminar, Cleveland State University, Cleveland, OH
- Sept 2020 **Invited Speaker**, *Mathematics Provides Insight into Self-Organization in Biology*, Junior Seminar Fall 2020, Cleveland State University, Cleveland, OH
- Sept 2020 **Invited talk**, *Mathematical Modeling of Foraging and Trail Formation in Ants*, Ecology, Evolution and Environmental Science Seminar, Cleveland State University, Cleveland, OH
- Sept 2020 **Invited Speaker**, *NSF GRFP: Knowing the Reviewers*, NSF GRFP Workshop, CSU Graduate School, Cleveland State University, Cleveland, OH
- Sept 2020 **Invited Speaker**, *NSF GRFP: Research Statement*, NSF GRFP Workshop, CSU Graduate School, Cleveland State University, Cleveland, OH
- Sept 2020 **Invited Speaker**, *NSF GRFP: Personal Statement*, NSF GRFP Workshop, CSU Graduate School, Cleveland State University, Cleveland, OH
- Sept 2020 **Invited Speaker**, *What is the NSF GRFP?*, NSF GRFP Workshop, CSU Graduate School, Cleveland State University, Cleveland, OH
- Aug 2020 **Invited talk**, *All About Me: Where Math Meets biology*, OpSTEM Cohort Welcome Seminar, Cleveland State University, Cleveland, OH
- May 2020 **Invited Poster Presenter**, *Module-based Learning and Open Textbooks in Linear Algebra*, 2020 Provost's Teaching Summit, Cleveland State University, Cleveland, OH
- Apr 2020 **Invited Speaker**, *Big Data and Modern Society*, College of Science Research Day Panel, Cleveland State University, Cleveland, OH
*Cancelled due to COVID-19 Outbreak
- Feb 2020 **Invited Speaker**, *Mathematics Provides Insight into Self-Organization in Biology*, Junior Seminar Spring 2020, Cleveland State University, Cleveland, OH

- Feb 2020 **Invited Speaker**, Mathematics Provides Insight into Self-Organization in Biology, SPS/Choose Ohio First Seminar, Cleveland State University, Cleveland, OH
- Feb 2020 **Invited Speaker**, Mathematics Provides Insight into Self-Organization in Active Biosystems, Biology Colloquium, Cleveland State University, Cleveland, OH
- Oct 2019 **Invited Speaker**, Effective Properties and Collective Dynamics in Bacterial Suspensions, Chemical and Biomedical Engineering Colloquium, Cleveland State University, Cleveland, OH
- Oct 2019 **Invited Speaker**, Applying for the NSF GRFP, Math Club, CSU, Cleveland State University, Cleveland, OH
- Sept 2019 **Invited Speaker**, Mathematics Provides Insight into Self-Organization in Biology, Junior Seminar Fall 2019, Cleveland State University, Cleveland, OH
- Sept 2019 **Invited Speaker**, NSF GRFP: Knowing the Reviewers, NSF GRFP Workshop, CSU Graduate School, Cleveland State University, Cleveland, OH
- Sept 2019 **Invited Speaker**, NSF GRFP: Research Statement, NSF GRFP Workshop, CSU Graduate School, Cleveland State University, Cleveland, OH
- Sept 2019 **Invited Speaker**, Why Open Education Resources and Student Impacts, Opener for Nicole Finkbeiner - Open Stax Resources, Library, Cleveland State University, Cleveland, OH
- Sept 2019 **Invited Speaker**, NSF GRFP: Personal Statement, NSF GRFP Workshop, CSU Graduate School, Cleveland State University, Cleveland, OH
- Sept 2019 **Invited Speaker**, What is the NSF GRFP?, NSF GRFP Workshop, CSU Graduate School, Cleveland State University, Cleveland, OH
- Sept 2019 **USRA Poster**, Active particles near an air/water interface, Fall USRA Poster Session, Cleveland State University, Cleveland, OH
- Sept 2019 **Invited Speaker**, Mathematical Modeling Provides Insight to Biology, Junior Undergraduate Seminar: CSU, Cleveland State University, Cleveland, OH
- Jul 2019 **Invited Speaker**, How To Do Great Research and Scholarship Pt. 2, CSU USRA Student Seminar, Cleveland State University, Cleveland, OH
- Jun 2019 **Invited Speaker**, How To Do Great Research and Scholarship Pt. 1, CSU USRA Student Seminar, Cleveland State University, Cleveland, OH
- Apr 2019 **Invited Speaker**, Applied and Computational Mathematics at CSU, College of Engineering April 2019 Meeting, Cleveland State University, Cleveland, OH
- Mar 2019 **Invited Speaker**, Mathematics Provides Insight into Self-Organization in Biology, Junior Seminar Spring 2019, Cleveland State University, Cleveland, OH
- Mar 2019 **Invited Speaker**, Mathematics Provides Insight Into Insecticide Effectiveness, Math Club: Cleveland State University, Cleveland, OH
- Oct 2018 **Invited Speaker**, Open Problems in Active Biosystems, Junior Undergraduate Seminar: Cleveland State University, Cleveland, OH
- Apr 2018 **Invited Speaker**, Results of Using OneNote to Enhance Learning in Calculus I-II, 2018 Provost's Teaching Summit, Cleveland State University, Cleveland, OH
- Mar 2018 **Invited Guest Speaker**, Using Microsoft OneNote to Improve Outcomes in MTH 181, EST 499: Cleveland State University, Cleveland, OH
- Feb 2018 **Invited Speaker**, Open Problems in Active Biosystems, Junior Undergraduate Seminar: Cleveland State University, Cleveland, OH

- Sept 2017 **Invited Speaker**, *Open Problems in Active Biosystems*, Junior Undergraduate Seminar: Cleveland State University, Cleveland, OH
- May 2017 **Invited Speaker**, *Using OneNote to Enhance Learning in Calculus I-II*, 2017 Provost's Teaching Summit, Cleveland State University, Cleveland, OH
- Feb 2017 **Invited Speaker**, *Open Problems in Active Biosystems*, Junior Undergraduate Seminar: Cleveland State University, Cleveland, OH
- Nov 2016 **Invited Speaker**, *Open Problems in Active Biosystems*, Junior Undergraduate Seminar: Cleveland State University, Cleveland, OH
- Oct 2016 **Invited Speaker**, *Collective Motion: Why 2 is Better Than 1*, Math Club: Cleveland State University, Cleveland, OH
- Sept 2016 **Invited Speaker**, *Collective Motion: Why 2 is Better Than 1*, Choose Ohio First/STEM Cohort Seminar: Cleveland State University, Cleveland, OH

Service

National

- National Science Foundation (NSF) Panel Reviewer (2019), (2021), (2024)
- 2021-Present Ohio MAA Project NExT Mentor (K. Erdem, U. Cinci)
- 2019-Present Reviewer for Mathematical Reviews (MathSciNet)
- 2018-Present Skype a Scientist (K-12 Outreach, 2018 8th Grade, 2019 Kindergarten, 2021 7th and 8th Grade, 2023 1st Grade)
- 2017-Present Greater Cleveland STEM Foundation Volunteer Resource
- 2017 Poster Judge, Assoc. of Women in Mathematics Poster Session (SIAM Annual Meeting 2017)
- 2017-2020 Abstract Reviewer for Ohio Academy of Sciences Yearly Conference

University/College Committees

- 2026-Present, Member, University Committee on Athletics
- 2025-Present, Member, Faculty Executive Committee for General Studies, College of Arts and Sciences
- 2025-Present, Member, Academic Steering Committee for College of Arts and Sciences
- 2024-2026, Member, Peer Review Committee (PRC), College of Arts and Sciences (Chair AY 25-26)
- 2024-2025, Chair/Member, Distinguished Faculty Awards Committee, CSU Faculty Senate
- 2024-Present, External Member, PRC-Tenure Committee (Dept. of Physics)
- 2020-Present, co-Director, CSU Center for Applied Data Analysis and Modeling (ADAM)
- 2022-Present, Member, CSU Provost's High Performance Computing Taskforce (HPC)
- 2022-Present, Member, University Open Education Resources (OER) Committee
- 2022-2024, Member, College of Arts and Sciences e-Learning Committee (Chair in AY 23-24)
- 2021-Present, Member, B.S. in Data Science Program Committee (Advisory Board)
- 2023, Reviewer for University Graduate Student Awards
- Spring 2022 Reviewer for CSU Outstanding Doctoral Research Award in Sciences and Math
- AY21-22, AY 22-23, Launch Team Faculty Mentor for S. Froehlich and D. Gao
- 2021-2022, Member, COSHP Curriculum and Instruction Committee
- 2021-Present, Member, CSU Academic Research Multisite Taskforce (replacing Academic server)
- 2021-2022, Member, COSHP-CLASS Workload Working Group - overseeing merging of college
- Fall 2021, co-Lead, COSHP-CLASS Study Circle for book "Noise".
- 2021 Gave a Center for Faculty Excellence Presentation on Using OneNote for an Engaged Classroom
- 2021 Participant in Honors College External Review Discussion as General Honors Course Instructor
- 2021 FRD Reviewer, CSU Office of Research
- 2019-2020, Faculty Advisor, Machine Learning for Science and Engineering Club (MLSEC)

- 2019-Present Annual Fall CSU NSF Workshop to prepare 12-15 students to submit NSF Graduate Research Fellowship Applications (presented 4 talks on strategy in applying and performed internal CSU review before apps were sent out)
 - 2025 Honorable Mention Kyra Stovicek (Mech Eng. Human Machine Systems)
 - 2023 Winner Ryan Godin (Chemistry)
 - 2021 Winner Shaye Tiell (Mech. Engin.)
 - 2 CSU Winners in 2020: 1. Niksa Praljak (Math/Physics) and 2. Farid Khoury (Chemical Engineering)
- 2019-2020 CSU Graduate Student Awards for Teaching Assistants Committee
- 2019-2020 Selwin Varghese (Advisor: C. Wirth), Ph.D. Committee - discontinued Spring 2020
- 2019 Search/Hiring Committee for Chem. and BioMed Eng. Postdoc (PI Wirth) - hired Sergio Dominguez-Medina
- 2018-2019, McNair (Non-Discipline Advisor) for David Mikhail (Engineering)
- Invited Speaker for EST 499 (Education Seminar, J. Kilbane)
- Fall 2018 Math Major Discussion Panel for COSHP by CSU SGA
- 2018-2019 CSU Graduate Student Awards for Teaching Committee
- 2018-2019 Jiarui Yan (Advisor: C. Wirth), M.S. Thesis Committee ('19), Ph.D. Committee
- 2017-Present Aidin Rashidi (Advisor: C. Wirth), Ph.D. Candidate
- 2017-2022 Level 1 University Graduate Faculty
- 2017-2018 CSU Thesis and Dissertation Award committee
- 2017 CSU-Faculty Scholarship Initiative (FSI) Proposal Reviewer
- 2016-2017, 2019 Student Selection Committees for NEOMED-CSU Partnership for Urban Health
- Faculty Representative at Fall 2016, Fall 2018, Spring 2020, Spring 2022, Fall 2024 Commencement

Departmental Committees

- 2026 Tenure-Track Mathematics Search Committee
- 2025-Present Undergraduate Program Director and Faculty Advisor for Math Majors and Math/Stat Minors
- 2024-Present Assessment Committee for Undergraduate Program B.S. in Math
- 2024-Present Undergraduate Program Committee
- 2022-Present Applied Math Committee (Chair '22-'24)
- 2021-Present Dept. Peer Review Committee-Tenure (Chair '23,'24,'25)
- 2017-Present Math Technology Coordinator
- 2017-Present Math Technology Committee (Chair '17-Present)
- 2022-2024 Tenure-Track Search Committee
- 2016-2024 Graduate Program Committee (Chair '20-'23)
- 2016-Present Colloquium Committee (Chair '16-Present)
- 2016-Present Social Media/Math Dept Webpage Committee (Chair of Committee/Webpage Editor)
- 2024-Present Point Person for Promotion of B.S./B.A. in Mathematics
- 2021-2022 Tenure-Track Search Committee for Applied and Computational Assistant Professor (Chair) hired A. Hoover in 2022
- 2021 Dept. of Mathematics and Statistics Chair Search Advisory Committee (Chair)
- 2018-2019 Tenure-Track Search Committee (Appointments/Hiring) hired Visiting Assistant Professor Dan Florentin in 2019
- 2016-2017, 2018-Present Calculus Committee

Editorial Board Member:

- 2025-Present, Associate Editor, *Involve: A Journal of Mathematics*
- 2025-Present, Review Editor, *Frontiers in Systems Biology* (Multiscale Mechanistic Modeling)

Referee For:

- 2026-Present Communications in Partial Differential Equations
- 2026-Present Ecological Modeling
- 2026-Present Materials
- 2025-Present IEEE Control Systems
- 2024-Present The ISME Journal (International Society of Microbial Ecology) IF: 11
- 2024-Present SIAM Multiscale Modeling and Simulation (MMS)
- 2024-Present La Mathematica (Journal of the American Association of Women in Math)
- 2024-Present SIAM Review (SIREV)
- 2022-Present Bioinspiration & Biomimetics
- 2021-Present PLoS Computational Biology
- 2021-Present Science Advances
- 2021-Present Physical Biology
- 2020-Present Transaction of the American Mathematical Society
- 2020-Present Royal Society Interface
- 2020-Present Applied Sciences, MDPI Journal
- 2019-Present Journal for Mathematical Biology, Springer
- 2019-Present Mathematics, MDPI Journal (ISSN 2227-7390)
- 2019-Present Mathematical Reviews, American Mathematical Society (2 reviews total)
- 2018-2019 Springer Book "Cell Movement: Modeling and Applications"
- 2018-Present Entropy, MDPI Journal, (ISSN 1099-4300, IF 1.947)
- 2017-2018 Book Reviewer for Mathematics & Computer Science Dept of CRC Press/ Taylor & Francis Group
- 2017-Present Journal of the Royal Society Interface
- 2017-Present Fluids, MDPI Journal, (ISSN 2311-5521, IF) (2 reviews total)
- 2017-2018 Book Reviewer for Elsevier (Mathematics, Composites and Construction Materials)
- 2014-Present SIAM Journal of Applied Mathematics (SIAP)
- 2014-Present Proceedings of the Royal Society A
- 2013-Present Mathematical Models and Methods in Applied Sciences (M3AS)
- 2013-Present Nonlinearity (IOP Publishing)

Peer Review Certificates:

- 2016 Springer Peer Review Academy Certificate - Springer Journals

External (Community) Service and Outreach:

- 2019-2025 (7x) NE Ohio Science Olympiad Event Coordinator/Judge (Codebusters Div B/Div C)
- Made 2022 Ohio State Competition Exam
- Presented a Career as a Math Professor for 1st Grade Class (Ms. Norris) in Hudson, OH
- Skyped with 1st Grade (Ms. Silcox) Class in Hudson, OH about Math/Science Careers
- July 2017 Poster Judge for SIAM Women in Math Poster Session
- April 2016 Project Judge for Choose Ohio First Poster Exhibition
- 2015-2016 NE Ohio STEM Science Fair Project Mentor (Hudson High School, Hudson, OH)
- 2015 Oakwood Middle School STEM Math/Physics Day at Kent LCI - demonstrated exciting math and physical phenomena for middle school students to enrich science education
- 2014-2016 Akron Science Expo - Volunteer KSU Math and Science table for middle/high school children.
- Fall 2015, Fall 2014, Speaker at Choose Ohio First Seminar for students in the STEM fields who grew up in Ohio and chose to remain in Ohio for higher education.

--- Professional Memberships

- Pi Mu Epsilon (Math Honorary). Member since 2007.
- Society for Industrial and Applied Mathematics (SIAM). Member since 2009.

SIAM Activity Groups: Life Sciences ('09-Present), Dynamical Systems ('19-Present), Data Science ('20-Present), Material Science ('09-'19), & Partial Differential Equations ('09-'18).

- American Mathematical Society (AMS). Member since 2009.
- Society for Mathematical Biology (SMB). Subgroups: 1. Methods in Biological Modeling, 2. Education, and 3. Population Dynamics, Ecology & Evolution, Member since 2013.
- Ohio Academy of Sciences (OAS). Member since 2016.
- Ohio Supercomputing Center Principal Investigator (PI). Member since 2017.
- International Liquid Crystal Society (ILCS). Member 2016-2018.

Student Research Advising Summary (since CSU Start 2016)

- 135 Total Student Research Projects (107 undergrad, 28 grad)
- 84 Math B.S. Senior Capstone Projects (14 Honors)
- 18 Math M.S. Exit Projects
- 10 Undergraduate Summer Research Awards (USRA)
- 14 Manuscripts with CSU student co-authors (13 CSU student lead authors)
- 15 Undergraduate Researchers Nominated for COSHP/Arts & Sciences/CLASS Valedictorian (2 won)
- 29 Students went on to Graduate Programs (12 M.S., 17 Ph.D)
- Mentored 4 NSF GRFP Winners and 1 Honorable Mention (Praljak '20, Khoury '20, Tiell '21, Godin '23, HM Stovicek '25)
- 1 REU Student

Graduate Research Advising

28. 2024-2025 Rosalia Motta, M.S. Exit Project, Topic: Herding Effects on Predator-Prey Systems
 - Started as a Pricing Analyst I at Progressive Insurance in May 2025
27. 2024-Present Ryhan Uddin, Ph.D. in EECS, Committee Member, Advisor: S. Kumar
 - 2026 Outstanding TA Award, College of Engineering
26. 2024 Arpit Gopani, M.S. Math Exit Project, Topic: Machine Learning for Differential Eqns
 - Started position as a Data Analytics Intern at IPL Global
 - Outstanding Math/Stat Graduate Student '25
25. 2023-Present, Vyshnavi Ciluveru, Ph.D. in Chemical Engineering (co-advisor w/ D. Munther, C. Kothapalli), Topic: Experiments for Bacterial Contamination in Poultry Processes
 - Contributed talk at Produce Safety Meeting in NC, July 2025
 - Contributed Poster at International Association for Food Protection Meeting, Cleveland, OH July 2025
24. 2023 Madison Kramig, M.S. Math Exit Project, Topic: TBA
 - Began as Adjunct Math Faculty Member at Lorain Comm. College in 2024
23. 2022-2023 Melanie Wroblewski, M.S. Math Exit Project, Topic: Optimal Scheduling
 - Predictive Planning and Insights Analyst at Sherwin-Williams
22. 2022-2023 Benjamin Kovacic, M.S. Math Exit Project, Topic: PDE models for Pollution in Rivers
 - 2022-2023 Outstanding Graduate Student, Dept. of Mathematics and Statistics
 - 2023 Research Data Scientist 2 at the Cleveland Clinic
21. 2022-2023 Molly Walsh, M.S. Math Exit Project, Topic: Modeling Intestinal Bacterial Growth
 - Starting in Fall 2024, Mathematics Teacher, Hawken High School (OH)

20. 2021-2022 Ayat Almamy, M.S. Math Exit Project, Topic: Disease Transmission with Waning Immunity
 - Starting in Fall 2026, Ph.D. in Mathematical Sciences at Arizona State Univ.
19. 2021-2022 Jacob Vitale, M.S. Chem Engin. Committee Member (Advisor: G. Ao)
 - 2022 Started as a Consultant Engineer at FM Global
18. 2021-2022 Neda Abdollahi Nohouji, Ph.D. Mech. Engineering, Committee Member, Research: Mechanical Stability of Knee Replacements
 - 2022 Started as Human Factors Engineer at Apple
17. 2021-2022 Sean Doherty, M.S. in Mechanical Engineering, Committee Member, Research: Predictions of Indentation Stiffness of Musculoskeletal Regions Using Ultrasound
 - 2023 Started as a Biomedical Data Specialist at CancerLinQ
16. 2020-2021 Asma Alshehri, M.S. Math Exit Project, Topic: Fourier Analysis for Diffusion Equations
15. 2020-2021 Joe Khoury, M.S. Chem E, Comm. Member, Research: Fabricating Nanotube fibers
 - Current Ph.D. Candidate in Chemical Engineering, Rice University
14. 2020-Present Nerion Zekaj, (w/ A. Resnick and D. Munther), Research: Modeling Kidney Tubules, Modeling Bacterial Contamination in Poultry Plants
 - Currently Ph.D. Candidate in Applied Mathematics at UNC-Chapel Hill
13. 2019-2021 Rodney Moore, M.S. Math Exit Project, Topic: Modeling Anthrax Inhalation
 - M.S. in Mathematics at Georgetown University
 - 2022 Bioinformatician at the FDA
12. 2019-2021 Tyler Rhoades, M.S. Math Exit Project, Topic: Medical Imaging
 - 2021 Research Data Scientist 2 at the Cleveland Clinic
11. 2019-2021 Justin Simmerer, M.S. Math Exit Project, Topic: PrEP use in HIV Prevention
 - 2025 Visiting Lecturer, Cleveland State University
10. 2018-2020 Mohammed Kalil, M.S. Chem. Engin. (w/ C. Wirth), Topic: Clustering Janus Cells
9. 2018-2020 Nick Baumgartner, M.S. Math Exit Project, Topic: Modeling Interacting Janus Particles
 - 2024 Started as Assistant Professor, Slippery Rock Univ.
 - Ph.D. Student, Dept. of Mathematics, Univ. of Wyoming starting 2020-2024
 - 2019-2020 CSU Excellent Graduate Student Teaching Award
 - Publication in *ACS Omega*, Two others submitted
8. 2018-2019 Ibtisam Alsharari, M.S. Math Exit Project, Topic: Modeling with Laplace Equation
7. 2018-2019 Majidah Alshammari, M.S. Math Exit Project, Topic: Numerical Analysis of Physical Waves
6. 2017-2018 Sabah Almazariqah, M.S. Math Exit Project, Topic: Fluid Dynamics at Low Reynolds Number
5. 2017-2018 Hamdah Alsharari, M.S. Math Exit Project, Topic: Fluid Dynamics at High Reynolds Number
4. 2017 Justin Flaherty, Summer Research, Topic: Boundary Layer Formation in Earth's Atmosphere
 - 2025-Present Postdoctoral Fellow, Ohio State University

- Ph.D. Student at Ohio State University (Physics) start Fall 2018 (Ph.D. 2024)
- 3. 2016-2017 Zach McCarthy, Topics: Microtubule Jamming and Poultry Chilling Process Models
 - Won Graduate Student of the Year for all of CSU
 - Ph.D. Student at York University (Canada) starting Fall 2017
- 2. 2015 B. Green (Physics Ph.D. student at Penn State), Topic: G-L Superconductivity
 - NSF Graduate Research Fellowship (PSU Physics), Ph.D. Candidate
- 1. 2014 M. Mizuhara (Math Ph.D. student at Penn State), Topic: Cell Motility
 - Associate Professor (Math and Stat) at The College of New Jersey

Undergraduate Research Advising

- 107. 2026-Present Kazutada Nakao, CSU Soft Matter REU Student (co-advised w/ C. Wirth), Topic: Active Matter Modeling
- 106. 2026-Present Elias Saab, USRA Research (co-advised w/ C. Kothapalli, D. Munther), Topic: Bacterial Cross-contamination in Poultry Processing
- 105. 2026-Present Vincent Palombo, USRA Research (co-advised w/ C. Kothapalli, D. Munther), Topic: Bacterial Cross-contamination in Poultry Processing
- 104. 2025-2026 Christina Mark (Flagler College), Honors B.S. in Mathematics and Biology, Topic: The Metastability of Ant Trails on Simulated Lattices
 - Fall 2026, Ph.D. Student at UNC-Chapel Hill
- 103. 2025-2026 Tori Lowe, B.S. in Math, Topic: Math Modeling of Hormonal Cycles in Women
- 102. 2025-2026 Logan Roberts, B.S. in Math/UTeach, Topic: Student Prep in Future Performance
 - Outstanding Student Teacher for AY 25-26 in the CSU Teach Program
- 101. 2025-2026 Hollis King, B.S. in Math/UTeach, Topic: Project-based Learning in Mathematics Classrooms
- 100. 2025-2026 Nghi Le, Honors B.S. in Math /UTeach, Topic: Modeling Parent Help on Student Performance
 - Nominated for College of Arts and Sciences Valedictorian, Spring 2026
- 99. 2025-2026 Hannah McLeod, Honors B.S. in Math, Topic: Predicting Gross Book Sales Revenue Based On Word-Of-Mouth
- 98. 2025-2026 Racheal Ina, B.S. in Math/UTeach, Topic: How Opinions of Math Affect Student Performance
- 97. 2025 Chanboramey (Mei) Vann, B.S. in Math, Topic: Incorporating Real-time Traffic Data in to PDE Congestion Models
 - Fall 2025, hired as a Financial Analyst I in the Enterprise Data and Insight Department at Sherwin-Williams in the Prediction Planning/AI Division
- 96. 2025-2025 Michael Marston, B.S. in Math, Topic: Multi-linear Regression Models for Squared-Up Bat Rates in MLB
- 95. 2025-2025 Sam Bostick, B.S. in Math, Topic: Math Mathematical Analysis for College Football Ranking

94. 2025-2025 Michelle Rollins, B.S. in Math, Topic: Modeling Multi-seasonal Breeding with Energy Storage
93. 2025-2025 LaiAnna Loucks, Honors B.S. in Math, Topic: Multi-linear Regression Models for Spotify Music Popularity
 - June 2026 started as Underwriting Assistant at McGowan Companies
92. 2025-2025 Dakota Cody, B.S. in Math, Topic: Optimization in Supply Chains
91. 2025-2025 Azrael Manross, B.S. in Math, Topic: Modeling Cancer Tumor Progression
90. 2025-Present Sarah Fergus, B.S. in Chem E., USRA: Bacterial Contamination in Poultry Production
89. 2025 Anthony Mastandrea, B.S. in Math, Topic: Modeling Effect of Daylight on Disease Prevalence
88. 2025 Simon Richard, B.S. in Math/CS, Topic: Learning in LLM Models for AI Development
 - Starting Ph.D. in Computer Science at Purdue Univ. in Fall '25
 - Outstanding Mathematics Major '25
87. 2024-2025 Vanessa Boyce, B.S. in Math, Topic: Predictive Modeling of Antibiotic Resistance
 - Nominated for CSU Arts and Sciences Valedictorian 24'-25'
 - '24-'25 CSU Sorority Member of the Year
86. 2024-2025 Grace Miller, Honors B.S. in Math/Physics, Topic: Modeling Atomic Deposition on Surfaces
 - Ph.D. Student in Particle Physics, CWRU, starting Fall 2025
 - Nominated for CSU Arts and Sciences Valedictorian '25
 - CSU Arts and Sciences Outstanding Senior Scholar '25
 - Outstanding Mathematics Major '25
 - Outstanding Physics Major '25
85. 2024 Max Burmeister, B.S. in Math, Topic: Capital Investment in Econ Growth Models
84. 2024 Manch Gamache, B.S. in Math, Topic: Effect of Permanent Death on Zombie Dynamics
83. 2024 Caroline Kilbane, B.S. in Math, Topic: Ecological Disease Spread
82. 2024-Present Jason Simon, USRA Project on Math Modeling of Bacterial Cross-contamination in Chilling Process
 - Nominated for CSU Arts and Sciences Valedictorian
 - Outstanding Mathematics Major '25
81. 2024 Chase Robinson, B.S. in Mathematics, Topic: Modeling Predicts Dynamic Pricing for NBA Games
80. 2023-Present Tanner Greene, Tatum Martinez, COF Topic: Modeling Small Pox Outbreak
79. 2023-2024 Michael Long, B.S. in Mathematics, Topic: Modeling Stock Prices
 - Fall 2024 started M.S. in Computational Finance at Carnegie Mellon Univ.
 - Summer 2025 Data Science Intern at Citizens Bank
 - 2024 Nominated for College of Arts and Sciences Valedictorian
78. 2023-2024 Rosalia Motta, Honors B.S. in Mathematics, Topic: Modeling Ecology with Decision Trees
 - 2022-2023 Outstanding MTH/STA Tutor

77. 2023-Present Caroline Kilbane, B.S. in Mathematics, Topic: Ecological Disease Spread
76. 2022-2023 Joshua Ashby, B.S. in Mathematics, Topic: Modeling Political Party Formation
 - 2022-2023 Outstanding Undergraduate Mathematics Major
 - Nominated for College of Science Valedictorian Spring 2023
 - College of Arts and Sciences Outstanding Scholar Spring 2023
75. 2022-2023 Halani Magbag, B.S. in Mathematics, Topic: Modeling Invasion of Vampires
74. 2022-2023 Alaa Yahya, B.S. in Mathematics, Topic: Modeling Perfect Basketball Shots
73. 2022-2023 Mahamoud Musleh, B.S. in Mathematics, Topic: Stochastic Differential Equations for Financial Models
 - Ph.D. Student in Mathematics at Florida State University
 - 2022-2023 Outstanding Undergraduate Mathematics Major
 - Nominated for College of Science Valedictorian Spring 2023
72. 2022-2023 Caitlin Gibson, B.S. in Mathematics, Topic: Identifying Sound Patterns in Frog Movement
 - 2022-2023 Outstanding Undergraduate Mathematics Major
 - Won Choose Ohio Award of Excellence for Research Poster at the COF Conference Spring 2023
 - Currently a Quantitative Analyst Associate, Fair and Responsible Banking team, KeyBank
71. 2022-2023 Corey Short, B.S. in Mathematics, Topic: Competition in Ant Colonies
 - Nominated for College of Science Valedictorian Spring 2023
70. 2022-2023 Paul DiVenti, B.S. in Mathematics, Topic: Model for Optimizing Fencing Matches
 - M.S. in Math Education at Towson University, Fall 2023
 - High School Math Teacher in MD
69. 2022-2023 , P. Clere, A. Jandric, J. G. Rodriguez, W. Turner et al. Choose Ohio First Project on *The Existence Of Planet Nine: Possible Locations and Effects of a Ninth Planet Beyond Neptune*
68. 2022 Aiden Demkee, B.S. in Mathematics, Topic: Inverse Problems from Linear Algebra
67. 2022 Anthony Laurienzo, B.S. in Mathematics, Topic: Social Media and Political Polarity
66. 2022 Melanie Wroblewski, B.S. in Mathematics, Topic: Predicting Box Office Revenue for Movies
 - 2022-2023 Outstanding MTH/STA Tutor
 - M.S. in Mathematics at CSU
65. 2022 Kayla Drager, B.S. in Mathematics, Topic: Modeling Genetically-Modified Crops
 - 2022-2023 Outstanding Undergraduate Mathematics Major
 - Named College of Arts and Sciences Outstanding Scholar Fall 2022
64. 2021-2022 Abbigail Walls, B.S. in Mathematics, Topic: Modeling Coral Reef Depletion
 - M.S. in Clinical Mental Health Counseling
63. 2021-2022 Oscar Barillas, B.S. in Mathematics, Topic: Transmission of Disease and Travel
62. 2021-2022 Kelly Cooper, B.S. in Mathematics, Topic: Projecting NBA Outcomes due to Injury
61. 2021-Present Zack Hatten, B.S. in Mathematics, Topic: Thermal Property Estimation of Fibrous Insulation: Heat Transfer Modeling and the Continuous Genetic Algorithm
60. 2021-2022 Rachel Schonhiutt, Honors B.S. in Mathematics, Topic: Modeling the Metabolism
 - Nominated for College of Science Valedictorian Spring 2022

59. 2021-2022 Julia Rausch, Honors B.S. in Mathematics, Topic: Data Analysis of Carbon Sequestration
58. 2021-2022 Benjamin Kovacic, Honors B.S. in Mathematics, Topic: Modeling Pollution Spread in Aquatic Ecosystems
 - Best Overall Presentation at 2022 Choose Ohio First Research Conference
 - M.S. in Mathematics at CSU
 - Outstanding Senior, College of Science and Health Professions Spring 2022
 - 2020-2021 B. Kovacic, Choose Ohio First
 - 2023 started as a Research Data Scientist 1 at the Cleveland Clinic
57. 2021 Evan Sadler, B.S. in Mathematics, Topic: Spread of COVID-19 on Networks
56. 2021 Daniel Boyer, B.S. in Mathematics, Topic: Game Theory in Baseball
55. 2021 Michael Miller, B.S. in Mathematics, Topic: Modeling Spread of COVID-19
 - Current: Quantitative Analyst at PNC
54. 2020-2022 Ryan Godin, B.S. Chemistry, Topic: Modeling Synthetic Cell Consortia
 - Currently Ph.D. Student in Chemical Engineering at Iowa State University (2022-)
 - Awarded NSF Graduate Research Fellowship 2023-2026
 - Outstanding Senior College of Science and Health Professions 2022
 - 2nd Place Poster Award in Food, Pharma, and Biotech at ASC21 in Boston 2021 (AIChE Annual Student Conference)
 - Elsevier Poster Prize at Soc. for Math Bio International Conference June 2021 (Cell and Developmental Biology)
 - REU in Synthetic Biology at Iowa State University (Summer 2021)
53. 2020-2021 Kiran Meyers, B.S. in Mathematics, Topic: Gravitation Force Geometry
52. 2020-2021 M. Dixon, C. Farrington, C. Blatnik, I. Ulle, Topic: Analyzing BPM in Music Genres
51. 2020-2021 S. Alfaro, K. Bailer, G. Caskey, A. Nacarato, Choose Ohio First: Quantifying Carbon Sequestered in Common Ohio Trees
 - S. Alfaro is currently a Data Science Researcher at Cleveland Clinic
 - S. Alfaro named 2022-2023 Outstanding MTH/STA Graduate Student
 - Best Poster Presentation, 2021 Choose Ohio First Conference
 - M.S. in Statistics at CSU
50. 2020-2021 N. McFarren, T. Moodry, T. Neece, M. Shehata, Choose Ohio First: Water Distribution and Quality Management in Cleveland, Ohio
49. 2020-2021 Kayla Ball, B.S. Mathematics, Topic: Modeling Social Networks
48. 2020-2021 Damian Dziubek, B.S. Mathematics, Topic: Modeling REM Sleep
47. 2020-2021 Anna Pavlovski, B.S. in Major Math/Chem E, Topic: Reaction Rates in Bacterial Growth
46. 2020 Russel George (w/ T. Heus), B.S. Physics, USRA Topic: Cloud Dynamics
45. 2020 Alex Sukennyk (w/ T. Heus), B.S. Physics/Mech. E., USRA Topic: Cloud Dynamics
44. 2019-2021 Ariana Chriss, Honors B.S. in Biology and Math, Topic: Modeling Chromosome Dynamics
 - Ph.D. Student in Computational Biology at Mount Sinai (New York)
 - 2021 Outstanding Senior in the College of Science and Health Professions

- 2021 Doretta Thielker Award (Outstanding Biology Senior)
- 43. 2019-2021 Alexis Brown, Honors B.S. in Biology, Topic: Algae Bloom Formation
 - 2021 Outstanding Senior in the College of Science
 - 2021-Present Research Intern at Holden Arboretum Foundation
 - Participant in REU at OSU Stone Laboratory, Summer 2020
 - TRIO (First-gen) Student
- 42. 2019-2020 Tyler Rhoades, B.S. in Math and Physics, Topic: Image Restoration Using Math
 - Salutatorian for College of Science and Health Professions for Fall 2020
 - Data Scientist at Cleveland Clinic
- 41. 2019-2020 Cameron Lathem, B.S. Mathematics, Topic: Predictive Modeling for Drag Racing
- 40. 2019-2020 Julia Janik, Senior Project, Topic: Math Modeling of Tympanic Membrane (Ear)
- 39. 2019-2020 Kevin Lannoch, Honors Senior Project, Topic: Econ. Modeling of Economy
 - Nominated for College of Liberal Arts and Social Sciences Valedictorian
 - Technical Sales Representative at Lincoln Electric (starting Summer 2020)
- 38. 2018-2019 Kenyon Ponzie, Senior Project, Topic: Fuel Efficiency due to Air Resistance
- 37. 2018-2019 Emilyann Moenich, Honors Senior Project, Topic: Predator-Prey Eqns
- 36. 2019 Steven Bengele, B.S. Chem and Biomed Engin., USRA Topic: Janus Cells
- 35. 2019 Justin Thompson, B.S. in Physics, USRA Topic: Cloud Dynamics
- 34. 2018-2020 Niksa Praljak, Honors B.S. in Math and Physics, Topic: Pulsatile Kidney Tubules (w/ A. Resnick)
 - 2020 NSF Graduate Research Fellow
 - Currently a Ph.D. student at U. Chicago (starting Summer 2020)
 - Outstanding Poster at CSU Science Research Day 2019
 - 2020 College of Science and Health Professions Valedictorian
- 33. 2018-2021 Theresa Lincheck, B.S. in Physics/Music, USRA Topic: Cloud Dynamics
 - 2021 College of Science Valedictorian
 - 2021 Ph.D. Student, Univ. of Oklahoma, School of Meteorology
- 32. 2018-2019 Luke Thornsberry, Choose Ohio First Topic: Auto-Driving Car Efficiency
- 31. 2018-2019 Christine Campbell, Jake Derkacs, Olivia Safady, Sebastian Solomon, Choose Ohio First Project, Topic: Spiral Bacterial Colonies
- 30. 2018-2019 Benjamin Kirkpatrick, Senior Project, Topic: Gene Regulation
- 29. 2018-2019 Masroor Butt, Senior Project, Topic: Disease and Travel
- 28. 2018-2019 Justin Simmerer, Senior Project, Topic: Effectiveness of PrEP
 - Currently an M.S. student in CSU Math Dept.
- 27. 2018-2019 Jacob Clark, Senior Project, Topic: Fourier Analysis in Biology
 - Currently a data analyst at AML Rightsource
- 26. 2018-2019 Alexis Bloor, Senior Project, Topic: Spread of Disease Between Species
 - Currently an Associate Financial Analyst at Northwestern Mutual

25. 2018-2019 Emily (Alec) DeBoard, Honors Senior Project, Topic: Wound Healing
24. 2018-2019 Wei Jia, Senior Project, Topic: Boundary Layers Effects in Clouds
 - M.S. Student and Ph.D. at University of Cincinnati (Data Science) starting Fall 2019
 - Graduate Student Intern at Proctor and Gamble, Spring 2021
23. 2018-2019 Olivia Maslanka, Honors Senior Project, Topic: Synchronization of Fire Flies
22. 2018 Mike Mesi, Senior Project, Topic: Modeling of Solar Panels
21. 2018 Lauren Blasinsky, Honors Senior Project, Topic: Traffic Flow
20. 2018 Jeremy Aufderheide, Senior Project, Topic: Fracture Mechanics of Crack Propagation
 - Currently a Consulting Developer at Equity Engineering
19. 2018-Present Marola Issa (w/ C. Wirth), Summer Research, USRA Topic: Active Particle Dynamics
 - Ph.D. Student in Chemical Engineering at CWRU
 - Named Future Leader in Chemical Engineering for the N.C. State Chem E Symposium
18. 2017-2018 Christine Campbell and Jake Derkacs, Choose Ohio First Project, Topic: Modeling Wind Turbines
 - Christine awarded Outstanding Sophomore in Engineering for 2017-2018
17. 2017-2018 Rachel Sutor, Joseph Wolf, Hayden Bartolovich, and Zachary Opperman, Choose Ohio First Project, Topic: Scallop Theorem
16. 2017-2018 Valerie Vidal, Senior Project, Topic: Modeling Type-II Diabetes
15. 2017-2018 Anthony Campagna, Senior Project, Topic: Predictive Analysis for Ranking MLB Teams
14. 2017-2018 David Zimmerman, Senior Project, Topic: Bifurcation Analysis
 - Currently M.S. in Mathematics at CSU
13. 2017-2018 Matthew Gebbett, Senior Project, Topic: Predictive Analysis for Ranking English Soccer Teams
12. 2017-2018 Aeisha Kangan, Senior Project, Topic: Modeling Econ. Growth in Developing Nations
 - M.S. student in Global Health at Georgetown University
11. 2017-2018 Jen Worthy, Senior Project, Topic: Math Models for Pedestrian Dynamics
 - M.S. student at Johns Hopkins and works for Teach For America
10. 2017-2018 Maddie Burns, Senior Project, Topic: Model for Disease Dynamics in Women's Health
9. 2016-2018 Nicky Baumgartner, Senior Project, Topic: Crab/Ant Interaction on Xmas Island
 - Currently an Assistant Professor at Slippery Rock University (PA)
 - M.S. in Mathematics CSU, Ph.D. in Mathematics U. Wyoming in 2024
 - Two papers submitted for publication while an M.S. in Math at CSU
8. 2016-2018 William Calabrese, Senior Project, Topic: What Determines the Shape of a Cloud
 - Currently, Metrology Engineer and Software Developer at CEIA
 - Ph.D. Student at Case Western Reserve University (Physics) starting Fall 2018
7. 2017-Present Nick Barron, Summer Research, Topic: Fractal Cloud Structures
 - Senior Radar Scientist at Aeolus Labs
 - 2024 Ph.D. in Meteorology at Penn State University

6. 2017 Vladimir Sworski, Summer Research, Topic: How High Does the Lower Atmosphere Go?
 - [Valedictorian, College of Science and Health Professions at CSU](#)
 - 2024 Ph.D. in Mathematics at Colorado State University
5. 2016-2017 Anthony Sulak, Independent Study, Topic: Optimal Shape of Water Towers
 - Research Project published in SIAM Undergraduate Research Journal (SIURO)
 - 2014 Ph.D. in Applied Mathematics, Michigan State Univ.
 - Currently, Risk Analytics Researcher, Nationwide Insurance
4. 2016-2017 Jalisa Richardson, Senior Project, Topic: Algae Growth in Biodiesel Fuel Production
 - Currently Senior Process Manufacturing Engineer
 - M.S. Student at UW-Madison (Engineering Data Analytics) Fall 2018
3. 2016 R. Dovishaw (Kent State, Individual Honors Work, Topic: Math Methods in Physics)
2. 2016 J. Sobieski (Kent State, Choose Ohio First Project), [Awarded Best Poster Presentation](#)
1. 2013 Q. Donahue, Topic: Homogenization)
 - 2019 Ph.D. in Mathematics from University of Pittsburgh

High School Student Research Advising

3. 2026-Present, Amogh Tripathi and Gavin Savett (Hawken High School w/ Teacher Molly Walsh)
2. 2020-2021, Peter (Alice) Soprunov, Topic: Modeling Iceberg Structures
 - 2021 started B.S. in Physics at University of Chicago
1. 2016-2017, Dorrie (Dan) Pharis (w/ T. Heus), Topic: Modeling Cloud Size Distribution
 - 2017 started B.S. in Physics at the Ohio State University

Teaching Honors

- | | |
|---------------------------|--|
| 2023-2024 | CSU Distinguished Faculty Award for Teaching , <i>For Outstanding Teaching and Student Mentorship, two awards for tenured/tenure-track over whole university</i> |
| 2023-2024 | Jearl D. Walker Outstanding Teaching Award , <i>For Outstanding Teaching and Student Mentorship in the College of Arts and Sciences</i> |
| 2025, 2022, 2020,
2018 | CSU Golden Apple Teaching Award , <i>CSU Alumni Association, Recognizes the contributions of outstanding faculty/staff members on the lives of students</i> |
| 2022-2023 | Provost's Faculty Merit Award , <i>For Outstanding Teaching</i> |
| 2022-2023 | OSC Campus Champions , <i>Ohio Supercomputing Center, Recognized as liason between CSU and OSC while promoting OSC resources to faculty and students</i> |
| 2020-2021 | Faculty Innovator Award , <i>CSU Center for Faculty Excellence, Invited to present a webinar on keeping students engaged during remote learning.</i> |
| 2023,2020,2019 | Nominated for Jearl D. Walker COSHP Teaching Award , <i>Recognizes an individual faculty member in the CSU COSHP for outstanding teaching</i> |
| 2019-2020 | Provost's Faculty Merit Award , <i>For Outstanding Teaching and Research</i> |
| 2017-2018 | Provost's Faculty Merit Award , <i>For Outstanding Teaching and Research</i> |
| 2017-2018 | Teaching Enhancement Award , <i>Cleveland State University</i> |
| 2017-Present | Level 1 University Graduate Faculty , <i>Cleveland State University</i> |
| 2016 | Kent State Department of Athletics Faculty Honor , <i>Kent State University, For being instrumental in student-athlete academic success</i> |

- 2012 **Department of Mathematics Teaching Award**, *Department of Mathematics, Pennsylvania State University*
- Fall 2011 **Promotion to Graduate Teaching Associate**, *Department of Mathematics, Pennsylvania State University*
- Fall 2011 **Teaching With Technology Certificate**, *Graduate School, Penn State University*

○ Nominated for PSU Department of Mathematics Teaching Award in 2011, 2012, and 2013

Teaching (Cleveland State)

Rating for "What is your overall evaluation of this instructor?" out of 5 Student Evaluations

- Spring 2026 **Math 396: Junior Seminar**, *Instructor*, **TBD**
- Fall 2025 **Math 401: Mathematical Modeling**, *Instructor*, **4.93/5.00**
- Spring 2025 **Math 396: Junior Seminar**, *Instructor*, **4.95/5.00**
- Fall 2024 **Math 401: Mathematical Modeling**, *Instructor*, **5.00/5.00**
- Spring 2024 **Sabbatical**
- Fall 2023 **Math 577: Numerical Methods**, *Instructor*, **4.88/5.00**
- Fall 2022 **Math 311: Numerical Analysis**, *Instructor*, **5.00/5.00**
- Fall 2022 **Math 286: Intro to Differential Equations**, *Instructor*, **5.00/5.00**
- Spring 2022 **Math 311: Numerical Analysis**, *Instructor*, **4.81/5.00**
- Fall 2021 **Math 401: Mathematical Modeling**, *Instructor*, **5.00/5.00**
- Summer 2021 **Math 286: Intro to Differential Equations**, *Instructor*, **4.84/5.00**
- Spring 2021 **Math 311: Numerical Analysis**, *Instructor*, **4.91/5.00**
- Fall 2020 **Math 401: Mathematical Modeling**, *Instructor*, **4.88/5.00**
- Fall 2020 **Math 181H: Honors Calculus I**, *Instructor*, **4.89/5.00**
- Summer 2020 **Math 286: Intro to Differential Equations**, *Instructor*, **4.92/5.00**
- Spring 2020 **Math 182H: Honors Calculus II**, *Instructor*, **4.94/5.00**
- Fall 2019 **Math 288: Linear Algebra**, *Instructor*, **4.82/5.00**
- Fall 2019 **Math 401: Mathematical Modeling**, *Instructor*, **4.81/5.00**
- Summer 2019 **Math 286: Intro. to Differential Equations**, *Instructor*, **5.00/5.00**
- Spring 2019 **Math 311: Numerical Analysis**, *Instructor*, **4.90/5.00**
- Fall 2018 **Math 181H: Honors Calculus I**, *Instructor*, **5.00/5.00**
- Fall 2018 **Math 401: Mathematical Modeling**, *Instructor*, **4.90/5.00**
- Summer 2018 **Math 288: Linear Algebra**, *Instructor*, **5.00/5.00**
- Spring 2018 **Math 493: Math Modeling II**, *Instructor*, **5.00/5.00**
- Spring 2018 **Math 311: Numerical Analysis**, *Instructor*, **5.00/5.00**
- Spring 2018 **Math 181: Calculus I**, *Instructor*, **4.94/5.00**
- Fall 2017 **Math 401: Mathematical Modeling**, *Instructor*, **5.00/5.00**
- Fall 2017 **Math 181H: Honors Calculus I**, *Instructor*, **5.00/5.00**
- Summer 2017 **Math 286: Intro. to Differential Equations**, *Instructor*, **4.75/5.00**
- Spring 2017 **Math 311: Numerical Analysis**, *Instructor*, **5.00/5.00**
- Spring 2017 **Math 182: Calculus II**, *Instructor*, **4.92/5.00**
- Fall 2016 **Math 401: Mathematical Modeling**, *Instructor*, **5.00/5.00**

Fall 2016 **Math 182: Calculus II**, *Instructor*, [4.77/5.00](#)

Teaching (Kent State)

Rating for "Overall, how would you rate your learning experience in this course" out of 5 Student Evaluations

Spring 2016 **Math 32052: Math Methods for Physical Sciences II**, *Instructor*, [4.90/5.00](#)

Fall 2015 **Math 32051: Math Methods for Physical Sciences I**, *Instructor*, [4.92/5.00](#)

- No Teaching Fall '14-Spr. '15, covered by NSF Grant #1212046 (PI X. Zheng, co-PI P. Palfy-Muhoray).

Teaching (Penn State)

Rating for "Rate the overall quality of the instructor" out of 7 on SRTE Student Evaluations

Spring 2013 **Math 220: Matrices/Linear Algebra**, *Instructor for 3 Sections*, [6.77/7.00](#), [6.66/7.00](#), [6.59/7.00](#)

Spring 2012 **Math 251: Introduction to ODEs and PDEs**, *Instructor*, [6.97/7.00](#)

Spring 2012 **Math 251Z: Introduction to PDEs**, *Instructor*, [6.83/7.00](#)

Spring 2011 **Math 41: Algebra and Trigonometry (Precalculus)**, *Instructor*, [6.63/7.00](#)

Fall 2010 **Math 251: Introduction to ODEs and PDEs**, *Instructor*, [6.83/7.00](#)

Teaching (University of Akron)

Rating for "Overall, I rate this instructor as excellent" out of 5 on IDEA Student Evaluations

Spring 2009 **Mathematics for Elementary School Teachers I**, *Instructor*, [4.8/5.0](#)

Fall 2008 **Intermediate Algebra**, *Co-coordinator and Instructor*, [4.8/5.0](#)

Spring 2008 **College Algebra**, *Instructor*, [4.6/5.0](#)

Fall 2007 **Intermediate Algebra**, *Instructor*, [4.9/5.0](#)

Computer Skills

Languages Python, Fortran, C, C++, HTML, $\text{\LaTeX}$, CUDA (GPU)

Platforms Mac OSX, Unix, Linux, Windows

Programs Matlab, Scilab, Mathematica, Maple, Grace, Xfig, Inkscape, Microsoft Office

Languages

- English (native language, U.S. Citizen)
- German (Elementary Proficiency in Reading, Writing, and Composition)